

CLASS 6 MATH

Mastery & Olympiad Workbook

Integers, Algebra, Geometry & 100 Critical Thinking Puzzles

Chapter 1: Integers & Number Systems

SECTION A: CORE CONCEPTS

1. Arrange the following integers in ascending order: -15, 8, -4, 0, -2, 12.

2. Evaluate the absolute value expression: $|-45| + |-12| - |-5|$.

3. Add the integers: $(-34) + (56) + (-12)$.

4. Subtract: $45 - (-18)$.

5. Multiply: $(-12) \times (-8) \times (5)$.

SECTION B: REAL WORLD APPLICATIONS

1. The temperature in San Jose at 6:00 AM was 12°C . By noon, it rose by 8°C , and by midnight, it dropped by 15°C . What was the temperature at midnight?

2. Mukesh's bank account had a balance of \$1,250. He withdrew \$450, then wrote a check for \$900. What is his current balance? (Indicate overdraft with a negative sign).

3. A submarine dives 250 meters below sea level. It then rises 85 meters. What is its new depth represented as an integer?

Chapter 2: Fractions, Decimals, & Ratios

SECTION A: CORE CONCEPTS

1. Simplify the ratio 144 : 108 to its lowest terms.

2. Convert 0.625 into a fraction and a percentage.

3. Divide the fraction $\frac{5}{8}$ by $\frac{3}{4}$.

4. Find 45% of 1,200.

5. If $x : 5 = 12 : 20$, find the value of x .

SECTION B: REAL WORLD APPLICATIONS

1. The ratio of mystery books to science books in the Milpitas library is 5:7. If there are 350 science books, how many mystery books are there?

2. Priyanka bought 3.5 kilograms of peaches from the orchard at \$4.20 per kg. She handed the cashier a \$20 bill. How much change should she receive?

3. Out of 80 packets of Crunchley makhana, 24 are the new spicy flavor. What percentage of the packets are spicy?

Chapter 3: Introduction to Algebra

SECTION A: CORE CONCEPTS

1. Write an algebraic expression for '7 less than twice a number y '.

2. Solve the linear equation for m : $m + 24 = 65$.

3. Solve for p : $4p - 8 = 32$.

4. Evaluate the expression $3x + 5y$ when $x = 4$ and $y = 2$.

5. If $x/6 = 15$, what is the value of x ?

SECTION B: REAL WORLD APPLICATIONS

1. Atharv has 'x' pencils. Anaya has 5 more pencils than Atharv. Together, they have 25 pencils. Write an equation and find out how many pencils Atharv has.

2. The Jigyasa educational app costs '\$m' to host on cloud servers per month. Over an entire year, the hosting cost is \$1,440. Write an equation and solve for m.

3. A box contains 'n' packets of snacks. If 5 identical boxes contain 150 packets in total, find the value of 'n'.

Chapter 4: Geometry & Mensuration

SECTION A: CORE CONCEPTS

1. Find the area of a triangle with a base of 15 cm and a perpendicular height of 10 cm.

2. What is the perimeter of a regular hexagon with a side length of 8 meters?

3. Two angles of a triangle are 55° and 65° . Find the measure of the third angle.

4. Find the circumference of a circle with a radius of 14 cm. (Use $\pi = \frac{22}{7}$).

5. Find the volume of a cuboid with length 8 cm, width 5 cm, and height 4 cm.

SECTION B: REAL WORLD APPLICATIONS

1. Mukesh uses an EGO Power+ mower on a rectangular lawn that is 25 meters long and 18 meters wide. What is the total area of the lawn he needs to mow?

2. The wheel of a stroller has a radius of 21 cm. How far will the stroller move forward in one complete revolution of the wheel? (Use $\pi = \frac{22}{7}$).

3. The perimeter of a square garden is 120 meters. What is the length of one side of the garden, and what is its area?

Multi-Concept Combined Practice

Level 1: Core Proficiency (Average Student) - SET 1

1. Simplify the ratio 45:60 to its lowest terms.

2. Evaluate the expression: $|-25| - |-10|$.

3. Solve the equation for x: $x + 18 = 42$.

4. Find the Area of a triangle with a base of 12 cm and a height of 8 cm.

5. Convert 0.375 into a fraction in its simplest form.

6. What is 20% of 450?

7. Subtract the integers: $-15 - (-8)$.

8. Find the Least Common Multiple (LCM) of 12 and 15.

9. Divide the fraction $\frac{3}{4}$ by $\frac{1}{2}$.

10. A car travels 180 km in 3 hours. What is its average speed?

Multi-Concept Combined Practice

Level 1: Core Proficiency (Average Student) - SET 2

1. Write an algebraic expression for '7 less than 3 times a number y '.

2. Evaluate $4x + 5$ when $x = 3$.

3. Find the perimeter of a regular pentagon with a side length of 6 cm.

4. Convert 2.5 hours into minutes.

5. Multiply the integers: $(-12) \times 5$.

6. If the ratio of boys to girls is 3:5 and there are 15 boys, how many girls are there?

7. Find the volume of a cuboid with length 5m, width 4m, and height 2m.

8. What is the Highest Common Factor (HCF) of 24 and 36?

9. A packet of Crunchley makhana costs \$4.50. How much do 8 packets cost?

10. Solve for m: $5m = 65$.

Multi-Concept Combined Practice

Level 2: Advanced Application (Top 10%) - SET 1

1. The sum of three consecutive integers is 72. Write an equation and find the integers.

2. Anaya scored 85, 90, 92, and 88 on her first four tests. What must she score on the fifth test to average 90?

3. Find the circumference of a circle with a radius of 14 cm. (Use $\pi = \frac{22}{7}$).

4. A jacket originally costs \$80. It is on sale for 15% off. What is the sale price?

5. Simplify using BODMAS: $24 \div (8 - 2) + 5 \times 3$.

6. The angles of a triangle are in the ratio 2:3:4. Find the measure of the largest angle.

7. Subtract $3 \frac{1}{4}$ from $7 \frac{1}{2}$.

8. Priyanka drives at 60 km/h for 2 hours, then at 80 km/h for 1 hour. What is her average speed for the whole trip?

9. If $4x - 7 = 17$, find the value of x .

10. Find the area of a parallelogram with a base of 15 cm and a perpendicular height of 10 cm.

Multi-Concept Combined Practice

Level 2: Advanced Application (Top 10%) - SET 2

1. Atharv bought 3 books that cost $\$x$ each, and a pen that costs $\$5$. He spent $\$26$ in total. Write an equation and solve for x .

2. Find the prime factorization of 120 using exponents.

3. A recipe calls for $\frac{2}{3}$ cup of sugar to make 12 cookies. How much sugar is needed for 30 cookies?

4. Evaluate the algebraic expression $3a^2 - 2b$ when $a = 4$ and $b = 5$.

5. What is the cost of fencing a rectangular park 50m long and 30m wide at the rate of \$8 per meter?

6. Convert the repeating decimal $0.333\dots$ into a fraction.

7. Two supplementary angles are such that one is twice the other. Find the angles.

8. If a train 200 meters long crosses a pole in 10 seconds, what is its speed in km/h?

9. Divide 15.6 by 0.12.

10. The area of a square is 144 sq cm. What is its perimeter?

Multi-Concept Combined Practice

Level 3: Logical Reasoning (Top 1%) - SET 1

1. If $\frac{x}{3} + 5 = \frac{x}{2}$, find the value of x .

2. A water tank is $\frac{2}{5}$ full. After adding 18 liters, it is $\frac{7}{10}$ full. What is the total capacity of the tank?

3. Find the smallest number which when divided by 6, 8, and 12 leaves a remainder of 3 in each case.

4. A digital clock shows 03:15. What is the exact angle between the hour and minute hands? (The hour hand moves past 3!).

5. If 5 workers can build a wall in 12 days, how many days will 3 workers take to build the same wall?

6. The price of an item is increased by 20%, then decreased by 20%. What is the net percentage change?

7. Find the sum of all prime numbers between 40 and 50.

8. A wheel has a diameter of 70 cm. How many revolutions will it make to travel 2.2 km? (Use $\pi = \frac{22}{7}$).

9. Simplify the complex fraction: $(\frac{1}{2} + \frac{1}{3}) / (\frac{1}{4} - \frac{1}{6})$.

10. I am a 2-digit number. The sum of my digits is 9. If you subtract 27 from me, my digits are reversed. Who am I?

Multi-Concept Combined Practice

Level 3: Logical Reasoning (Top 1%) - SET 2

1. Mukesh invests \$500 at a simple interest rate of 4% per year. How long will it take for his investment to grow to \$600?

2. In a class, the ratio of boys to girls is 4:5. If 2 boys leave and 2 girls join, the ratio becomes 1:2. How many students were originally in the class?

3. Find the value of $1 - 2 + 3 - 4 + 5 - 6 \dots + 99 - 100$.

4. A solid cube of side 6 cm is painted blue on all outside faces, then cut into smaller 1 cm cubes. How many 1 cm cubes have exactly two blue faces?

5. Evaluate: $(2^5 \times 3^2) / (4^2 \times 9)$.

6. The average age of 5 friends is 12. If a new friend joins, the average age becomes 13. How old is the new friend?

7. A pipe can fill a pool in 4 hours, and another can empty it in 6 hours. If both are opened, how long will it take to fill the pool?

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8. Find the number of diagonals in a regular decagon (10-sided polygon).

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9. If $3a = 4b = 5c$, find the ratio $a : b : c$.

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10. A rectangular piece of paper 44 cm long and 10 cm wide is rolled along its length to form a cylinder. What is the volume of the cylinder? (Use $\pi = \frac{22}{7}$).

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Multi-Concept Combined Practice

Level 4: Critical Thinking (Top 0.1%) - SET 1

1. Solve the cryptarithm: $CROSS + ROADS = DANGER$. Given that $R=6$ and $S=3$. (This might be too long, let's substitute: $SEND + MORE = MONEY$, what is O ?)

2. Find the last two digits of 3 to the power of 40.

3. You have a 5-liter jug and a 3-liter jug. How do you measure exactly 4 liters of water?

4. A frog is at the bottom of a 30-meter well. Every day it climbs 3 meters, but slips down 2 meters at night. On which day does it escape?

5. How many zeros are there at the end of 100 factorial (100!)?

6. Two trains, 200 km apart, are traveling towards each other at 50 km/h. A bird flies back and forth between them at 75 km/h until they crash. How far did the bird fly?

7. If you multiply all the odd numbers from 1 to 99, what is the units digit?

8. How many triangles are there in a pentagram (a 5-pointed star)?

9. A man has 53 socks: 21 blue, 15 black, 17 red. How many must he pull out blindly to ensure a matching pair?

10. Using exactly four 4s and any math symbols, create an equation that equals 20.

Multi-Concept Combined Practice

Level 4: Critical Thinking (Top 0.1%) - SET 2

1. You have 9 identical-looking coins. One is fake and slightly heavier. Using a balance scale exactly twice, how do you find the fake coin?

2. Three friends check into a \$30 hotel room, paying \$10 each. The clerk realizes the room is \$25 and gives \$5 to the bellboy to return. The bellboy keeps \$2 and gives \$1 to each friend. They paid \$9 each (\$27) + \$2 bellboy = \$29. Where is the missing dollar?

3. Complete a 3x3 magic square with numbers 1-9 so all rows, columns, and diagonals sum to 15.

4. If 1.5 cats catch 1.5 mice in 1.5 minutes, how many cats catch 60 mice in 60 minutes?

5. A hollow cube has an external volume of 1000 cm^3 . The walls are 1 cm thick. What is the volume of the empty space inside?

6. Find the sum of the interior angles of a 12-sided polygon.

7. I am a 3-digit number. My hundreds digit is twice my tens digit. My ones digit is the sum of the other two. What are the possible numbers?

8. If you divide 100 by half and subtract 10, what do you get?

9. A grandfather clock takes 6 seconds to strike 4 o'clock. How many seconds will it take to strike 8 o'clock?

10. Evaluate the alternating harmonic series sum conceptually or logic puzzle:
What comes next? 1, 11, 21, 1211, 111221...

Multi-Concept Combined Practice

Level 5: National Olympiad (Top 10) - SET 1

1. There are 5 houses of 5 colors. 5 people of different nationalities live there, drink different beverages, smoke different cigars, and have different pets. If the Brit lives in the red house... (What is the name of this famous logic puzzle?)

2. Find the greatest common divisor of 111,111 and 11,111,111.

3. You are in a hallway with 100 lockers. You toggle every locker, then every 2nd, then every 3rd... up to 100. Which lockers remain open?

4. A spider web doubles in size every day. It covers a window in 30 days. On what day was the window half covered?

5. How many 3-digit numbers contain the digit '5' at least once?

6. Four people cross a bridge at night with one flashlight. They take 1, 2, 5, and 10 minutes. Max 2 people can cross at a time. What is the shortest time for all to cross?

7. A king has 1000 wine bottles, 1 is poisoned. The poison takes 24 hours. He has 10 prisoners. How can he find the poisoned bottle in 24 hours?

8. What is the maximum number of pieces you can cut a circular pizza into using 4 straight planar cuts?

9. If A, B, C are distinct non-zero digits and $AAA + BBB + CCC = BAAC$, find A, B, C.

10. How many total rectangles of any size can you count in a 4x4 grid?

Multi-Concept Combined Practice

Level 5: National Olympiad (Top 10) - SET 2

1. The sum of the digits of a 2-digit number is 10. If 18 is added to the number, the digits reverse. What is the number?

2. A monkey climbs a 60 ft pole. He goes up 3 ft in one minute, and slips down 2 ft the next minute. How many minutes to reach the top?

3. Two cars go from A to B. Car 1 does half distance at 30mph, half at 60mph. Car 2 does half time at 30mph, half time at 60mph. Who arrives first?

4. You have a 7-minute hourglass and an 11-minute hourglass. How do you boil an egg for exactly 15 minutes?

5. What day of the week will it be exactly 10^6 days from today, if today is Thursday?

6. There are 25 horses. You can race 5 at a time, no stopwatch. Minimum races to find the top 3 fastest horses?

7. If you multiply all numbers on a phone keypad, what is the result?

8. A magical lily pad doubles every day. If it takes 48 days to cover the pond, when did it cover a quarter of the pond?

9. If A=1, B=2... Z=26, what 4-letter word has the product of its letters equal to 455?

10. Find a 10-digit number where the first digit tells how many 0s are in the number, the second tells how many 1s... up to 9s.
